

Md Faisal Kabir, M.sc.

🏠 Schönberger Str. 76, 24148 kiel.

✉️ mfk@informatik.uni-kiel.de

🌐 <https://www.linkedin.com/in/md-faisal-kabir-2203a4a1/>

🆔 <https://orcid.org/my-orcid?orcid=0009-0000-2354-2193/>

Personal profile

Hands-on experience in medical and clinical data analysis, data structuring, and prediction and classification modelling. Strong technical background in Python, SQL, and data formats, combined with practical experience in interdisciplinary academic and industrial environments.

Employment History

- 2022 – Present 📌 **Research Associate**, Intelligent Systems Group, Kiel University, Kiel
Research focus:
- Working on medical and clinical data for basic analysis, data structuring, and data flow management
 - Predictive modeling for process optimization
 - Classification of diseases, medications, and measurements in diverse research datasets
 - Managing and curating research data from clinical and biomedical studies
 - Implementing data integration pipelines based on text, image, tabular, time-series etc.
 - Ensuring data quality, consistency, and traceability across projects
 - Supporting scientific publications and research documentation
 - Collaborating with interdisciplinary partners in medicine, informatics, and data science
 - Machine learning and data-driven modeling focus on multimodalities
 - Predictive modeling for process optimization
 - KIT2 (TeleNotArtz) project completed successfully
- 2020 – 2021 📌 **Data Scientist**, Research and Development Department, Paul Wurth S.A., Luxembourg
Work focus:
- Data collection, and validation for industrial projects
 - Predictive modeling and process optimization using machine learning
 - Ensuring data integrity and quality assurance
 - Collaboration with international research and engineering teams
 - Application of deep learning and statistical inference for process improvement

Education

- 2022 – present 📌 **PhD student, Christian-Albrechts-Universität zu Kiel (CAU)** , Intelligent Systems, Department of Computer Science.
Research focus: AI-supported data management and integration of clinical data for prediction and classification models.
- 2016 – 2021 📌 **M.Sc., Fachhochschule kiel**, Information Engineering.
Thesis title: Usage and optimization of CycleGAN for multivariate time series analysis.
- 2009 – 2014 📌 **B.Sc. American International University - Bangladesh**, Electrical and Electronic Engineering.
Thesis title: American sign language image processing using matlab to communication with hearing impaired.

Skills

Data Management	SQL/MySQL, CSV, Excel, Data Cleaning, Quality Assurance, LaTeX.
Programming & Tools	Python, R, JavaScript, NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch, Keras, OpenCV, Git, Docker, Jupyter etc.
Visualization Tools	Matplotlib, Seaborn, Tableau, Power-BI.
Cloud Service	Experience in Azure cloud and Data Factory.
Languages	English – Fluent German – B1 (B2 in progress).

Research Publications

Journal Articles

- 1 M. F. Kabir, M. M. I. Nayem, and S. Tomforde, "Enhancing disease classification in emergency medical scenarios through synthetic data: A comparative study," *Lecture Notes in Computer Science (LNCS)*, 2025.

Conference Proceedings

- 1 M. F. Kabir, Nayem, and S. Tomforde, "Diagnosis and treatment support as ai-driven assistant system for telenotary emergency care," in *Proceedings of the 18th International Conference on Agents and Artificial Intelligence - ICAART, INSTICC, SciTePress*, 2026.
- 2 M. F. Kabir, M. M. I. Nayem, and S. Tomforde, "Synthetic data generation for emergency medical systems: A systematic comparison of tabular gan extensions," in *Proceedings of the 17th International Conference on Agents and Artificial Intelligence - Volume 3: ICAART, INSTICC, SciTePress*, 2025, pp. 1199–1206, ISBN: 978-989-758-737-5. DOI: 10.5220/0013307200003890.
- 3 M. F. Kabir and S. Tomforde, "A deep analysis for medical emergency missing value imputation," in *Proceedings of the 16th International Conference on Agents and Artificial Intelligence - Volume 3: ICAART, INSTICC, SciTePress*, 2024, pp. 1229–1236, ISBN: 978-989-758-680-4. DOI: 10.5220/0012457300003636.
- 4 M. F. Kabir et al., "Organic computing," S. Tomforde and C. Krupitzer, Eds., kassel university press, 2024, ch. 10, pp. 121–135. DOI: 10.17170/kobra-202402269661.

References

Prof. Dr. Ing. Sven Tomforde
Department of Intelligent Systems
Kiel University
Tel: +49 431 880-4844
st@informatik.uni-kiel.de